

CNNs Revisited: What Transformers Taught Convolution

Applied Deep Learning — Chapter 18

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Foundation Books Project

Roadmap

From ViTs Back to ConvNets

What Modern CNNs Keep

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Fine-Tuning ConvNeXt on Food-101

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Summary

From ViTs Back to ConvNets

Why Revisit CNNs After ViTs?

Vision Transformers changed how we model images, but they did not retire convolution.

- Convolution still encodes locality, weight sharing, and translation equivariance — useful priors for natural images.
- ConvNeXt asks: *which Transformer-era choices help a ConvNet, and which were never about attention at all?*
- Many “Transformer wins” came from training recipes, not the architecture name.

The contest is not CNN vs. Transformer; it is which recipe earns the next validation point.

Running Example: Food-101

- Closed-set image classification: 101 classes, 101,000 images.
- Carve a validation split from the official training data; reserve the official test split for the selected recipe.
- Approximate working split: 60,600 train + 15,150 val + 25,250 test (official).
- Top-1 and top-5 accuracy are both reported because dishes are visually close (ravioli, gnocchi, dumplings).

Goal: a reproducible classroom baseline, not a leaderboard score.

What Modern CNNs Keep

Inheritances From Classical CNNs

ConvNeXt still keeps:

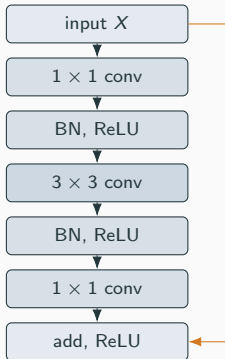
- spatial feature maps and shared kernels
- stride, padding, downsampling
- locality and translation equivariance

Why these survive: textures, edges, grill marks, and noodle strands appear anywhere on a plate. Local shared detectors are the right prior. The new question is not how convolution works — it is what changed around it.

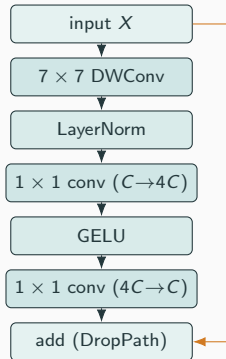
The ConvNeXt Block

ConvNeXt Block vs. ResNet Block

ResNet block

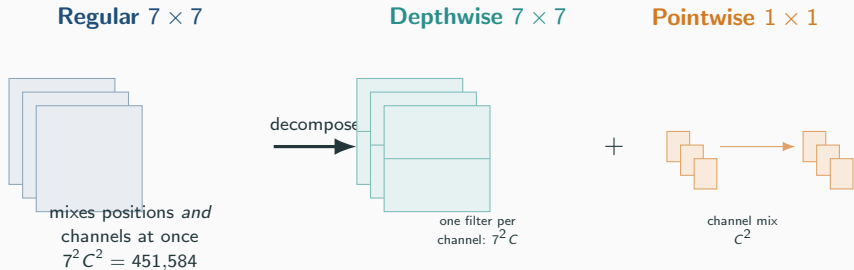


ConvNeXt block



Positions mix once (DWConv), channels mix once (1×1 expansion), then add.

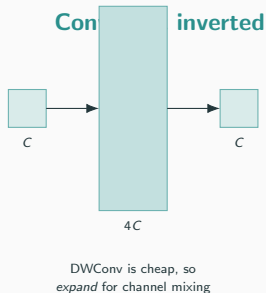
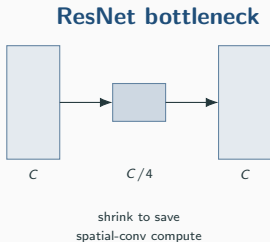
Why Depthwise + Pointwise?



$C = 96$: $451,584 \rightarrow 7^2 \cdot 96 + 96^2 = 13,920$ weights. Cheaper spatial mixing makes **large kernels practical**.

Caveat: fewer parameters $\neq$ less wall-clock time.

Inverted Bottleneck



Capacity spent where useful: combining low-level cues into class evidence. For Food-101: “toasted surface + circular shape + sauce” is one channel-mixer decision.

The Transformer-Era Recipe

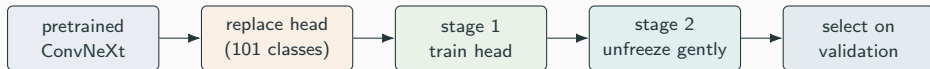
ConvNeXt's gains are not all in the block — they include the training recipe:

- LayerNorm instead of BatchNorm
- GELU instead of ReLU
- AdamW with decoupled weight decay
- stochastic depth on residual branches
- stronger augmentation, longer schedules

Change any of these and you are running a different experiment. A bare “ConvNeXt” label is not a recipe.

Fine-Tuning ConvNeXt on Food-101

Fine-Tuning Workflow



- Frozen-head stage debugs labels, shapes, and head training cheaply.
- Cautious unfreezing adapts features without destroying them.
- Test set is touched once, on the selected recipe.

Head Cost is Small

Pretrained body output dimension 768 $\rightarrow$ Food-101 head:

$$768 \cdot 101 + 101 = 77,669$$

trainable weights and biases.

Tiny compared with the full ConvNeXt body — which is exactly why training the head first is a fast plumbing check.

Augmentation should preserve labels: random crops if food remains visible, horizontal flips usually fine, vertical flips usually not.

Reference Run (ConvNeXt-Tiny on Food-101)

Setting	Value
Checkpoint	TorchVision ConvNeXt-Tiny IMAGENET1K_V1
Image size / resize	224×224 / 256
Optimizer	AdamW, weight decay 0.05
Stage 1 LR / Stage 2 LR	3×10^{-4} / 10^{-5}
Schedule	1 frozen-head epoch + 4 fine-tune epochs
Hardware / time	classroom CUDA, $\approx$ 20.6 minutes
Frozen-head val top-1	63.97%
Selected val top-1	78.36% (epoch 4)
Final test top-1 / top-5	83.52% / 96.61%

A reproducible baseline, not a leaderboard score.

MetaFormer: It's the Block, Not the Mixer

Reshape an $H \times W \times C$ feature map to $N \times C$ tokens.

- **Token mixer:** moves information across the N positions (attention, conv, pooling, MLP).
- **Channel mixer:** transforms the C -dim feature at each position.

Most modern blocks separate these two jobs. Recognizing this prevents “architecture-name thinking.”

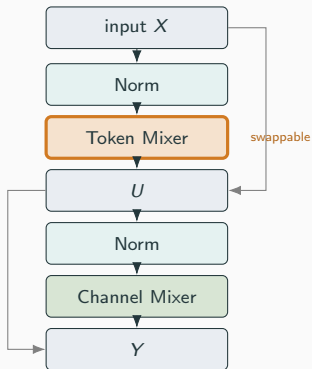
The MetaFormer Block

$$U = X + \text{TokenMixer}(\text{Norm}(X))$$

$$Y = U + \text{ChannelMixer}(\text{Norm}(U))$$

Token mixer is swappable: attention (ViT), conv (ConvNeXt), pooling (PoolFormer), token MLP (MLP-Mixer).

PoolFormer's surprise: even *pooling* keeps the block competitive — the surrounding recipe carries much of the weight.



Mixer Comparison

Mixer	What it moves	Caution
Convolution	Local neighborhood, shared weights	Global context arrives only with depth or large kernels
Self-attention	Many or all tokens, input-dependent	Quadratic cost in token count
Pooling	Nearby values by a fixed rule	Cheap; less adaptive
Token MLP	Learned position-mixing	Sensitive to input size and data scale

Compare the whole block, not just the mixer's name.

Large Kernels and Long Range

Why Make the Kernel Big?

Texture alone does not always identify a dish. Plate-level arrangement may be needed. Routes to wider context for ConvNets:

- more layers (depth)
- more downsampling (resolution loss)
- **larger kernels** via depthwise efficiency

31×31 convolution, $C = 96$:

- regular: $31^2 \cdot 96^2 = 8,856,576$ weights
- depthwise: $31^2 \cdot 96 = 92,256$ weights

RepLKNet showed large depthwise kernels can close part of the ViT gap.

Effective Receptive Field; Shape vs Texture

Effective receptive field The region that *actually* influences a feature in practice — often smaller than the nominal one.

Texture bias Classify by local pattern.

Shape bias Classify by global form.

Food-101 contains both. Sushi, ramen, cheesecake, ice cream — texture and arrangement both matter. **No bias is universally better.** Inspect errors and let validation decide.

ConvNeXt V2 and Self-Supervised ConvNets

Masked Modeling Meets Convolution

ConvNeXt V2 co-designs self-supervised pretraining and architecture:

- Fully convolutional masked autoencoder (FCMAE) reconstruction loss

$$\mathcal{L}_{\text{FCMAE}} = \frac{1}{|M|} \sum_{j \in M} \|x_j - \hat{x}_j\|_2^2$$

- Global Response Normalization inside the block.

Lesson: pretraining objective and architecture are not independent. A “pretrained ConvNeXt” is not one model.

Checkpoint Provenance Matters

Two ConvNeXts can share architecture but differ in:

- training data (ImageNet-1K vs 22K vs FCMAE)
- objective (supervised vs masked-image)
- resolution and schedule
- license

Symptom on Food-101: one checkpoint helps sushi, another helps soups. Before declaring a model-family winner, name the **exact builder + weights**.

Choosing a Backbone

The Resource Budget Is a Metric

Backbone choice is not just accuracy.

Situation	First model	What to measure
Natural-image classification, mid-sized data	Pretrained CNN / ConvNeXt	val acc, runtime, memory, errors
Strong ViT checkpoint, long-range cues matter	Pretrained ViT or hier-ViT	gain vs token cost
Detection / segmentation	Dense-prediction backbone	task metric, latency
Classroom hardware, uncertain pipeline	Small pretrained model first	whether the pipeline is correct

Cost is a metric. Report latency, throughput, and memory with accuracy.

Reporting and Debugging

Debugging Order on Food-101

Symptoms suggest different first moves:

- Frozen head poor $\Rightarrow$ check labels, paths, head shape.
- Frozen head fine, fine-tune degrades $\Rightarrow$ LR, augmentation, BatchNorm behavior.
- Train acc 98%, val acc 76% $\Rightarrow$ overfit / aug too weak.
- Both low $\Rightarrow$ undertrained, wrong LR, label parsing bug.

Tiny overfit test on a few batches is a cheap, powerful first check.

What the Confusion Matrix Tells You

Off-diagonal entries point to:

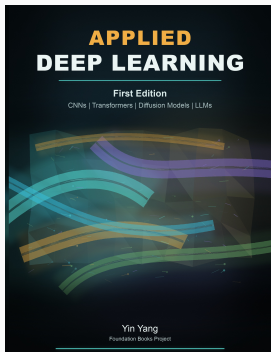
- visually similar classes (ravioli / gnocchi / dumplings)
- one popular label dominating (often a training or label bug)
- systematic mix-ups worth a closer look

Companion artifact: a CSV of (image path, true, predicted, confidence) — not embedded photos. Renders are local where licenses allow.

Summary

- Convolution remains useful: locality, sharing, equivariance fit images.
- ConvNeXt modernizes ResNet-style ConvNets with depthwise + pointwise separation, inverted bottleneck, LayerNorm, GELU, AdamW, stochastic depth.
- MetaFormer: most of the win is in the block, not just the mixer.
- Large depthwise kernels and self-supervised ConvNets (V2) extend the family further.
- Choose by validation evidence and resource budget — name the checkpoint, not just the architecture.

Next: vision-language models — when the output becomes language conditioned on an image.



Applied Deep Learning

Chapter overview slides for the book by Yin Yang.

Book page	foundation-books.github.io/applied-deep-learning
Code	github.com/foundation-books/applied-deep-learning-code
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